

RESEARCH ARTICLE

A Systems Engineering Analysis of Three-Point and Four-Point Wind Turbine Drivetrain Configurations

Y. Guo¹, T. Parsons^{1,2}, K. Dykes¹, R. N. King^{1,3}

¹National Renewable Energy Laboratory, 15013 Denver West Parkway, Golden, CO 80401. ²Colorado School of Mines, Department of Mechanical Engineering, 1500 Illinois Street, Golden, CO 80401. ³University of Colorado at Boulder, Department of Mechanical Engineering, 1111 Engineering Drive, UCB 427, Boulder, CO 80309.

ABSTRACT

This study compares the impact of drivetrain configuration on the mass and capital cost of a series of wind turbines ranging from 1.5-MW to 5.0-MW power ratings for both land-based and offshore applications. The analysis is performed with a new physics-based drivetrain analysis and sizing tool, Drive Systems Engineering (DriveSE), which is part of the Wind-Plant Integrated System Design & Engineering Model. DriveSE uses physics-based relationships to size all major drivetrain components according to given rotor loads simulated based on International Electrotechnical Commission design load cases. The model's sensitivity to input loads that contain a high degree of variability was analyzed. Aeroelastic simulations are used to calculate the rotor forces and moments imposed on the drivetrain for each turbine design. DriveSE is then used to size all of the major drivetrain components for each turbine for both three-point and four-point configurations. The simulation results quantify the trade-offs in mass and component costs for the different configurations. On average, a 16.7% decrease in total nacelle mass can be achieved when using a three-point drivetrain configuration, resulting in a 3.5% reduction in turbine capital cost. This analysis is driven by extreme loads and does not consider fatigue. Thus, the effects of configuration choices on reliability and serviceability are not captured. However, a first order estimate of the sizing, dimensioning and costing of major drivetrain components are made which can be used in larger system studies which consider trade-offs between sub-systems such as the rotor, drivetrain and tower. Copyright © 0000 John Wiley & Sons, Ltd.

KEYWORDS

Systems engineering; drivetrain design; sensitivity analysis; cost optimization

Correspondence

Y. Guo, National Renewable Energy Laboratory, 15013 Denver West Parkway, Golden, CO 80401. E-mail: yi.guo@nrel.gov

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1. INTRODUCTION

Three-point and four-point suspensions, which refer to wind turbine drivetrain configurations with either one or two main bearings respectively, are the most common wind turbine drivetrain architectures. In the three-point suspension configuration, the rotor is rigidly connected to the main shaft, which is supported by a single main bearing near the rotor. A shrink disk typically connects the downwind side of the shaft to the low-speed stage of the gearbox. The gearbox is supported by two torque arms that are connected to the bedplate elastically. These two torque arms, along with the single main bearing, provide a total of three points of support. Commercial wind turbines that utilize this configuration include the General Electric GE 1.5-MW, Siemens SWT108 2.3-MW, Nordex N117 2.4-MW, and Vestas V112 3-MW [1]. Four-point suspension configurations, sometimes referred as two-main-bearing suspension configuration, place an additional main bearing near the down-wind side of the main shaft with the intent of isolating any nontorque rotor loads upwind of the gearbox. Nontorque rotor loads are the non-torsional loads transmitted from the rotor blades to the drivetrain. These nontorque loads can affect gearbox reliability significantly by causing uneven loads among planetary gears and reducing bearing life [2]. The design protects the gearbox from nontorque loads but, at the same time, such designs can be sensitive to deflections and tolerances in the assembly [1]. Turbines that utilize this configuration include the Gamesa G114 2 MW, Vestas V80 2 MW, and General Electric GE120 2.5 MW.

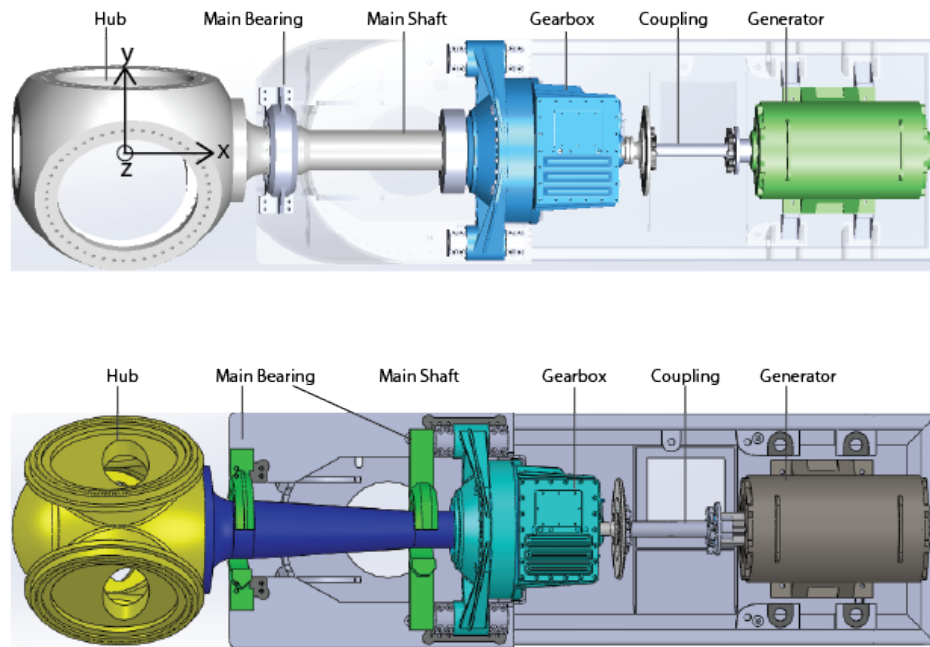


Figure 1. Three-point (top) and four-point (bottom) architectures [3].

The choice of one or two main bearings is a primary driver of nacelle component sizes, masses, and costs. Nacelle mass directly affects the structural and dynamical requirements of the turbine tower, foundation or offshore platform, and even the crane used for land-based installation. Consequently, main-bearing configuration is a critical design choice that affects overall wind turbine and plant cost of energy. This paper presents a new modeling toolset for and comparative analysis of three- and four-point wind turbine drivetrain configurations for two open-source turbine designs in land-based and offshore loading conditions.

Publicly available, peer-reviewed literature comparing the design trade-offs associated with these two drivetrain architectures is scarce. Industry reports have claimed that four-point suspensions are more costly by having the additional main bearing, but that the designs have advantages in the isolation of nontorque loads from the gearbox [4, 5]. Test results published through the Gearbox Reliability Collaborative (GRC) [6], which uses a three-point suspension drivetrain in its analysis, indicate that such nontorque loads should not affect tooth contact patterns in the intermediate and high-speed stage in a manner that would decrease gearbox life. At the same time, however, those loads may significantly affect planet-ring gear mesh pattern and planet-carrier misalignment in the low-speed stage [2]. The resultant edge loading, they concluded, may contribute to high contact stresses and shorter gear life [2]. Understanding the true effects of loading on the gearbox and how to prevent failures is needed to create an accurate model that accounts for known failure modes. While a gearbox design that considers these complex failure modes associated with nontorque loading is outside the scope of this study, the analysis here shows the impact of the configuration on the rest of the drivetrain design with additional qualitative discussion on the potential impacts to the gearbox.

This drivetrain analysis also serves as a demonstration of the capabilities of a new systems engineering modeling tool for wind turbines and plants and the value of integrated design and analysis approaches. Work in this field began with a University of Sunderland study in 1993, which defined a method for approximating the size and cost of key components in horizontal axis wind turbines (Sunderland model) [7]. The Sunderland model consists of a set of semi-empirical formulations for drivetrain components by estimating design loads at the rotor, propagating these loads through the entire system, and using the loads to estimate the size of each component. The Sunderland model was calibrated to data on actual turbines of that time [7, 8]. Following the Wind Partnerships for Advanced Component Technology (WindPACT) work that occurred between 2002 and 2005 [9], the National Renewable Energy Laboratory (NREL) released a new wind turbine design Cost and Scaling Model (NREL CSM) that utilized the work from the University of Sunderland and updated the cost and size formulations to better reflect modern turbine technology [8].

While the NREL CSM improved the overall cost estimation for larger turbines, it abstracted away from the engineering analysis foundations of the original Sunderland model and detailed design studies performed under WindPACT. Seeing the need for physics-based models that are sensitive to loading and physical design considerations, the NREL wind energy systems engineering group developed and released a series of physics-based models for a wind turbine as part

of Wind-Plant Integrated System Design & Engineering Model (WISDEMTM) [10]. WISDEM integrates wind turbine, wind plant, and various costs models to allow for integrated wind turbine and plant optimization. The WISDEM model is compatible with the Framework for Unified System Engineering and Design of Wind Plants (FUSED-Wind) that has been jointly developed by NREL and the Technical University of Denmark. Both WISDEM and FUSED-Wind are built around the National Aeronautics and Space Administration's open-source Multidisciplinary Design Analysis and Optimization (OpenMDAO) tool, which is a modular optimization tool that interconnects individual simulation codes for system-level studies. For more information on the software and its functionalities, detailed documentation can be found online [10]. The DriveSE model was developed as an important model within WISDEM to enable investigations of various design changes on design, sizing and cost of the turbine. As an example, DriveSE has been applied in full turbine optimizations that investigate trade-offs across sub-systems including the rotor, drivetrain and tower. [11, 12].

In this study, we demonstrate the DriveSE capability through an examination of the trade-offs associated with three-point and four-point drivetrain designs using the WISDEM model. Because the WISDEM drivetrain module, DriveSE, sizes drivetrain components using rotor loading from an upstream model, model outputs such as component geometry, mass, and cost properties can be more closely linked to rotor loads of a given blade design and even site selection (e.g., a land-based or offshore location) [3]. This integrated analysis aids in evaluating design choices such as main bearing configurations that may strongly affect overall turbine capital costs. This work will explore the impact of the drivetrain configuration design at both the component mass and system cost levels to understand the key trade-offs in the system in both a physics- and cost-based analysis framework.

2. WISDEM DRIVETRAIN MODEL

DriveSE consists of a series of interacting mathematical models of drivetrain subcomponents as shown in Figure 2 and described in detail in the DriveSE theory manual [3]. At this time, DriveSE contains physical loads-based models for the low-speed shaft, main bearings, gearbox, bedplate, and yaw system. The remainder of the components in the hub and nacelle systems are sized using scaling relationships based on empirical data. Master routines in DriveSE interface with other wind turbine components of WISDEM, the rotor and tower, but also with the turbine and plant cost models. At this top level, design criteria on allowable stress, deflection, and nacelle center of gravity are applied as constraints to an optimization of each sub-component for the minimum weight. This is an improvement over the previous systems engineering drivetrain models mentioned earlier [7, 8], which used empirically based scaling relationships that became outdated as technology evolved.

Key model inputs include the extreme aerodynamic rotor loads (torque and nontorque), gravity loads, gearbox configurations such as the number and type of gear stages, and design parameters such as rotor overhang and gearbox location. The outputs of DriveSE falls into two categories: subcomponent outputs and system outputs. Subcomponent outputs include the dimension and weight of individual subcomponents, as well as the gearbox stage ratios and stage masses. The mass outputs for all the individual components are then used in a turbine cost model as part of overall system cost analysis. The system outputs are the cumulative weight, moment of inertia, and center of gravity of the entire hub and nacelle assemblies, which are used as inputs at the tower design level as well as inputs to plant cost models where those system attributes impact installation costs. The individual drivetrain component models in WISDEM were compared against known component specifications as well as higher-order finite element analysis [3]. Results from different models agree reasonably well.

Several general assumptions have been made to facilitate drivetrain modeling in DriveSE. The current modeling effort focuses on component designs driven by maximum stress and deflection criteria. By using safety factors recommended by the industry standard [13], this approach sizes components to avoid catastrophic failure. The deflection constraints [14] ensure components are properly aligned and within geometrical limits for bearing alignment, gear-tooth meshing, etc. These geometrical requirements are an important part of the integrated design work flow connecting main bearings, low-speed shaft (LSS), and gearbox components. Other general assumptions include homogeneously distributed component masses for calculating moments of inertia and center of mass (CM). For more information on the DriveSE model formulation, validation, and other documentation, refer to the DriveSE model report [3].

Each component model uses a unique design approach, and various assumptions were made for individual components. The hub is modeled of scaling arguments calibrated to industry data. The current implementation treats the hub as a thin-walled ductile cast-iron cylinder with circular holes for blade root openings and LSS flange. The hub outer dimensions and thickness scale to the rotor diameter and blade root thickness.

The main shaft and bearing are sized first by determining the length from deflection limitations imposed by main bearings, with a maximum length constrained by the rotor overhang distance. Distortion energy failure theory is used to determine the outer diameters at the main bearings, with the final design consisting of a hollow shaft with a taper or cylinder between bearing locations. The shaft shape is automatically determined by DriveSE. Main bearings are then selected based

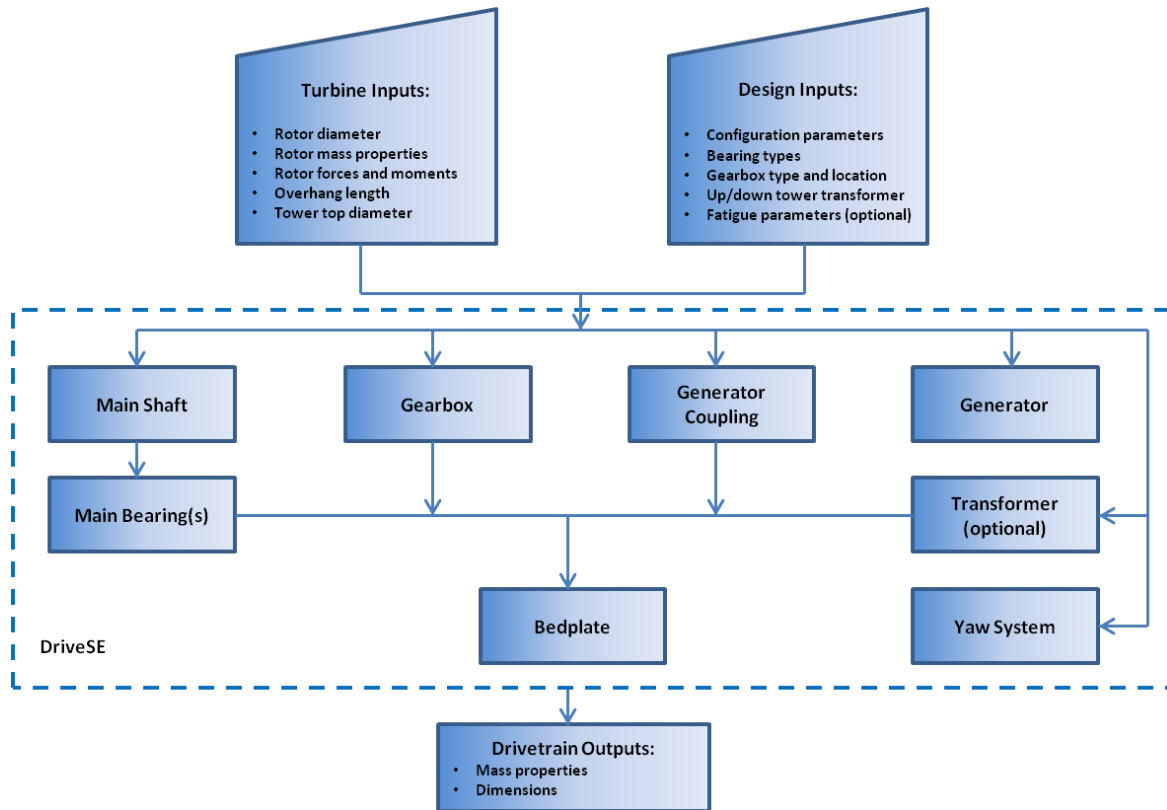


Figure 2. DriveSE calculation flowchart.

on shaft geometry and load capacity from the DriveSE database. Finally, the shaft dimensions are updated to match the closest bearing bore diameters.

The model designs the gearbox for the minimum weight by optimizing the speed ratio of each stage. The model considers gearboxes with two or three stages with different combinations of planetary and parallel stages. The gearbox model outputs the weight, volume, and speed ratio of each gearbox stage as well as the overall gearbox weight. Detailed dimensions of gearbox internal components are not provided at this time. Total efficiency of the turbine drivetrain (mainly gearbox and generator) is assumed as 95% in DriveSE.

The bedplate size is approximated by modeling the bedplate as two parallel I-beams and separately treating the upwind and downwind sections. The upwind section is assumed to be made of ductile cast iron; the downwind section is assumed to be made of steel. Weights of the drivetrain components are applied to the bedplate structure at the CM of each component, and rotor aerodynamic loads are superimposed on the upwind bedplate section. The upwind and downwind bedplate sections are individually sized to meet deflection and bending stress constraints that are automatically defined inside the program. Torque is currently not considered in the bedplate model because of its minor effects on the cost estimate.

A friction plate yaw bearing at the nacelle tower and several yaw motors constitute the yaw system. The friction plate bearing is treated as a steel annulus and is sized empirically according to the tower top diameter and rotor diameter. The motors are assumed to be flange-mounted and planetary gear systems. The number of motors is a function of the rotor diameter if not specified by the user.

The rest of the drivetrain and hub system is included in the analysis for sizing the bedplate and to compute the overall drivetrain weight, inertia, and CM. Masses and sizes of these components are currently calculated using a modified version of DriveWPACT [15], with updated CM calculations based on component sizes and CM optimization. The DriveWPACT models for the coupling, brake, and generator are highly simplified parametric models. A physics-based generator and generator-coupling model is not included at this stage, although a parametric model of the transformer has been added.

3. METHODS

Each component in DriveSE is sized based on a multitude of input parameters that specify the desired drivetrain configuration, the extreme rotor forces and moments, and the torque at rated power. Table I shows the required external or internal input variables for each component model.

3.1. Loads Sensitivity Analysis

Table I. External and internal input variables used in DriveSE component sizing

Input variable	main shaft	main bearing(s)	Gearbox	Bedplate
Torque at rated power (external)	✓	✓	✓	-
Power rating (external)	✓	✓	✓	-
Gearbox ratio (external)	-	-	✓	-
Rotor weight (external)	✓	✓	-	✓
Aerodynamic rotor loads (internal)	✓	✓	-	✓
Overhang distance (external)	✓	✓	-	✓
Tower-top diameter (external)	-	-	-	✓

Uncertainty involved in the aerodynamic rotor loads makes an understanding of individual component sensitivity to these loads valuable for further comparison between turbines. In addition, the drivetrain design's sensitivity to certain rotor loads can indicate the effects of the rotor design on the drivetrain weight and cost. A load's sensitivity analysis is performed to investigate the impact a change in load magnitude has on the mass of individual components. A base set of loads that included forces and moments is taken from a 5-MW land-based reference turbine [16] that will be discussed in further detail in the following section. The coordinate system used in this study is illustrated in Figure 1. A multiplier is added to each load individually, and the impacts on component mass in both three-point and four-point configurations are plotted. The bedplate and LSS are the only components analyzed in this study because the main bearing models are directly coupled to the shaft model, and the gearbox size is determined from the rated torque, which is a significantly less-uncertain input variable. This analysis yields a quantitative assessment of the effects of a wide range of load magnitudes on the DriveSE model results. The base loads used in sensitivity analysis are included in Table II, and were approximated from simulated loads found in the appendix for the 5-MW land-based reference turbine.

Table II. Base loads used in sensitivity analysis

Fx (kN)	Fy (kN)	Fz (kN)	Mx (kNm)	My (kNm)	Mz (kNm)
255	-180	-1,300	5,000	-14,000	-5,500

3.2. Drivetrain Case Study Methodology and Reference Turbines

After quantifying the sensitivity of drivetrain components to changes in the input loads, an analysis of three-point and four-point main bearing configurations was carried out for two reference wind turbine designs: the WindPACT 1.5-MW Turbine [17] and the NREL 5-MW reference turbine [16]. The key characteristics of each turbine are provided in Table III.

The WindPACT 1.5-MW Turbine is the result of a study on how new technologies and larger rotors would affect the cost of energy. Similar to the NREL 5.0-MW Reference Turbine, the WindPACT 1.5-MW turbine is a three bladed, upwind, variable speed, variable pitch conventional wind turbine design. Details on the WindPACT 1.5-MW are also included in Table III. The NREL 5-MW reference turbine is a conventional utility-scale turbine with a three-bladed upwind, variable-speed, variable-pitch design. It is loosely based on the REpower 5M, RECOFF [18], and Dutch Offshore Wind Energy Converter Project [19] designs and is a representative design for offshore turbines of a similar power rating. It is commonly used as a baseline design for wind energy research on diverse topics such as hydrodynamics of floating turbines [20], blade design [21], large-eddy simulations [22], extreme and fatigue-load studies [23, 24], and fluid-structure interactions [25]. Relevant geometric and mass properties for the NREL 5-MW reference turbine and its drivetrain are given in Table III, all of which are taken from Jonkman, 2009 [16].

These designs are chosen because they are publicly available and can provide a complete description of the blade and tower structural design along with sufficient information about the hub and drivetrain characteristics needed to perform aeroelastic simulations. Table IV provides a summary of the different simulation cases considered in this study. A unique set of loads was generated for each case by running a suite of simulations using NREL's aero-servo-hydro-elastic multibody

Table III. Wind turbine specifications

Parameter	1.5-MW WindPACT	5.0-MW NREL Reference
Nameplate (kW)	1,500	5,000
Rotor diameter (m)	70	126
Hub height (m)	84	90
Cut-in, rated, cut-out wind speed (m/s)	4, 16.18, 25	3, 11.4, 25
Cut-in, rated rotor speed (rpm)	8.5, 20.5	6.9, 12.1
Gearbox ratio	1:97	1:78
Overhang distance (m)	3.3	5.0
Shaft tilt (degrees)	5.0	5.0
Hub center to main bearing along shaft (m)	1.535	1.912
Low-speed shaft mass (kg)	3,420	16,898
Tower-top diameter	2.30	3.78
Rotor mass (kg)	32,016	110,000
Nacelle mass (kg)	52,839	240,000
Bedplate mass (kg)	10,140	51,933
Hub mass (kg)	15,104	56,780
Tower mass (kg)	122,522	347,460

dynamic simulator, FAST [26]. The resulting loads were post-processed to determine the extreme loads for each design case using a MATLAB-based Generator of Extreme-Event Tables [27]. The design load cases and the resulting extreme loads for each simulation case are presented in the appendix.

Table IV. Summary of simulation cases

Case	Turbine	Drivetrain Design	Location
1	1.5-MW WindPACT	Three-point	Land-based
2	1.5-MW WindPACT	Four-point	Land-based
3	5.0-MW NREL reference	Three-point	Land-based
4	5.0-MW NREL reference	Four-point	Land-based
5	5.0-MW NREL reference	Three-point	Offshore
6	5.0-MW NREL reference	Four-point	Offshore

3.2.1. Component Sizing and Comparison

Because the design load cases (DLCs) are extreme loads and not expected to occur simultaneously, the masses of each component were calculated for individual DLC load cases listed in the appendix independently. The governing load case was taken to be the one that produced the maximum component mass and may be different for each part. This approach was chosen as a conservative means of sizing the overall set of drivetrain components. In addition to rotor loads, the bedplate model takes into account the individual component weights that the bedplate must support. Because of this, a second iteration of the analysis was used in which DLCs were applied to the bedplate, with the maximum masses from each other component as manual inputs. The final assembly mass, including all above-yaw components that do not depend on turbine loading, is summed to arrive at a total nacelle mass figure.

The current version of DriveSE seeks to minimize the forces on the main bearings and reduce nontorque moments on the gearbox. This results in the main bearings for the four-point drivetrain being spaced as far apart as possible without exceeding main-shaft deflection constraints. However, the gearbox model does not currently account for the effects of nontorque loads on the gearbox design. For this reason, the study does not capture the trade-off between a longer and more massive main shaft in a four-point configuration versus a larger gearbox that would result from higher nontorque loads in a three-point design. Because this design trade-off is very important for a comparison between drivetrain configurations, the mass of the three-point gearboxes is adjusted to account for the bearing and housing mass increase that is normally accounted for in design.

The following process is followed to account for the mass difference in the gearbox:

1. Radial and axial loads on the gearbox are calculated in the shaft model, and it is assumed that two cylindrical roller bearings carry equally the nontorque loads at the low-speed stage of the gearbox.

2. Bearings whose load ratings can accommodate half of the radial load each, and with a bore diameter equal to the shaft diameter at the gearbox connection, are selected from SKF bearing catalogs [28].
3. The gearbox housing mass increase is taken to be directly proportional to the incoming loads on the gearbox, and scaled to fit available industry data.
4. The increase in bearing mass and housing mass within the gearbox is then summed to account for the total mass increase.

The individual component and overall subsystem masses for each case are then compared to explore the impact of the drivetrain configuration choice.

3.2.2. Component Costing and Comparison

At a system level, the design of the drivetrain predominantly affects the cost of individual components and turbine capital costs. To analyze the cost effects of drivetrain configuration on the capital cost of each component, masses resulting from calculations described in the previous section are connected to the inputs of the WISDEM Turbine Costs Systems Engineering model, which approximates component costs from scaling relationships between mass and cost [29]. Due to nonlinearities in the empirical data on mass-cost relationships, the cost comparison results differ from those that consider mass only. The cost results, as well as how they relate to the sizing results, are discussed for individual components.

4. RESULTS

4.1. Loads Sensitivity

Results of the loads sensitivity analysis on a three-point suspension drivetrain are shown in Figures 3 and 4. As the former figure conveys, the three-point main shaft model is highly sensitive to the nontorque moments M_y and M_z . These loads contribute to stress concentrations at the upwind main bearing, such that the shaft diameter at this location must be increased to sustain these larger bending moments, and the mass of the resultant tapered shaft is especially sensitive to these two moments. The moment in the y-direction has the largest impact on this model because its already high magnitude is compounded by the rotor overhang moment. In much the same way, M_z and F_y will compound their effects, but due to their lower magnitudes they are shown to have less of a total effect on the mass of the shaft. The main shaft mass is also dependent on the torque (M_x), which contributes to a torsional shear stress throughout the shaft, but has a relative significant effect on the stress at the smaller-diameter, downwind end of the component. The forces in the x- and y-directions are shown to have a nearly negligible effect on the mass of the main shaft because they do not contribute to the stress concentrations and deflections that drive the LSS sizing.

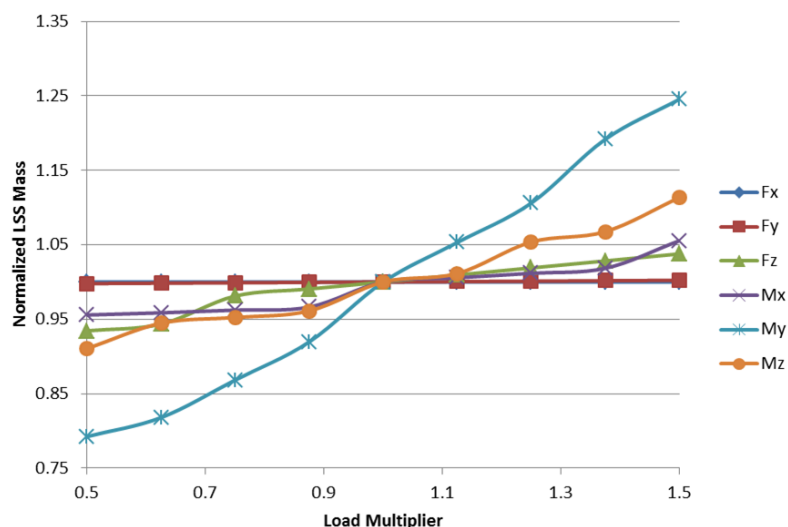


Figure 3. LSS loads sensitivity under three-point configuration

In Figure 4, the bedplate model is also shown to be highly sensitive to the moment in the y-direction for two distinct reasons: first, the same compounding effects of moment and complementary force, M_y and F_z impact the stress on the

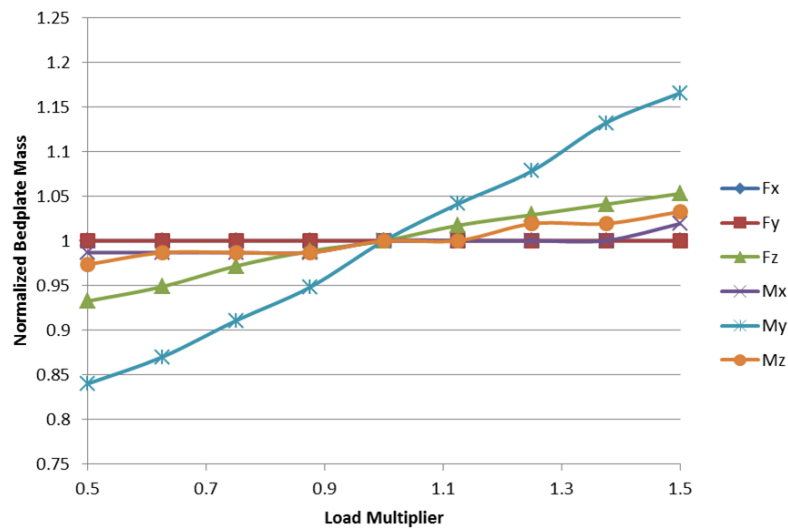


Figure 4. Bedplate loads sensitivity under three-point configuration

bedplate's upwind support structure. Second, the weight of the main shaft and bearings is an internal input to the bedplate model, such that the more massive shaft/bearing assembly weighs down the bedplate more and necessitates a larger and more massive support. In this way, the effects of this moment will always compound to impact the bedplate sizing. The only other load with a direct effect on the bedplate model is the force in the z-direction, which contributes to the bending stress along the upwind portion of the support. All other loads, to which this model is expectedly shown to have less sensitivity, only indirectly impact bedplate mass because of their moderate effects on other component weights.

Figures 5 and 6 show the same analysis performed with a four-point suspension drivetrain model. The main difference between these configurations is the existence of a downwind main bearing, which helps to carry forces and moments from the rotor. The main shaft results still show that the moments M_y and M_z have the largest effect on this component's mass. However, the model's sensitivity to each moment is almost the same, as opposed to the previous configuration which was much more sensitive to the moment in the y-direction. Under the three-point configuration, it is assumed that the gearbox trunnion force in the y-direction is negligible such that the moments are not carried completely through the shaft. Because of this, the rotor moments are less directly carried in the shaft without a second main bearing. In the four-point configuration, however, both moments are assumed to be carried by the main bearings such that the model's sensitivity to each load is nearly the same. The difference in the four-point sensitivity for the LSS between these moments arises mainly from the fact that component weights also affect the bending stress from M_y , so its effects normalized to the baseline loads are slightly smaller.

After normalizing against the mass of the components designed under the base loads, the slopes of these lines indicate that the four-point main shafts are less sensitive to loads. The upwind and downwind diameters between the main shaft models are similar because the stress concentrations on which the shaft diameter depends at these locations are nearly the same. The shaft under a four-point configuration is significantly longer than its three-point counterpart because the shaft deflection is much less in the four-point configuration than the three-point configuration. Only when the deflection constraints are not met does the model shorten the main shaft according to loading [3]. This leads to a higher sensitivity of the shaft mass to length, which is most often determined from variables other than loading. Consequently, in these four-point cases, loads have less of an effect on the mass of the main shaft than the maximum shaft length. It is important to note that the same trends in loads sensitivity are shown in the main bearing of the three-point suspension machine because the bearing model is closely coupled to the shaft diameter.

For the four-point shaft model, the bedplate model is still more sensitive to M_y . This is because, although M_z affects the weight of the shaft and bearings to be supported by the bedplate, the bedplate model does not directly consider stiffness in the y-direction to be a driver for the bedplate design. Much like in the three-point design, F_z and M_y , in addition to the component weights, are the only loads considered in the bedplate model. This is important to note because in several cases the DLC that produces the largest moment in the y-direction will determine the size of the bedplate, while the DLC with the largest combined moment produces the largest main shaft. Indeed, from the 5-MW land-based loads cases found in the appendix, the load instance with the highest moment in the y-direction (row 10) determined the bedplate size while the instance with the maximum combined moment (row 11) at the main bearing location determined the shaft sizing.

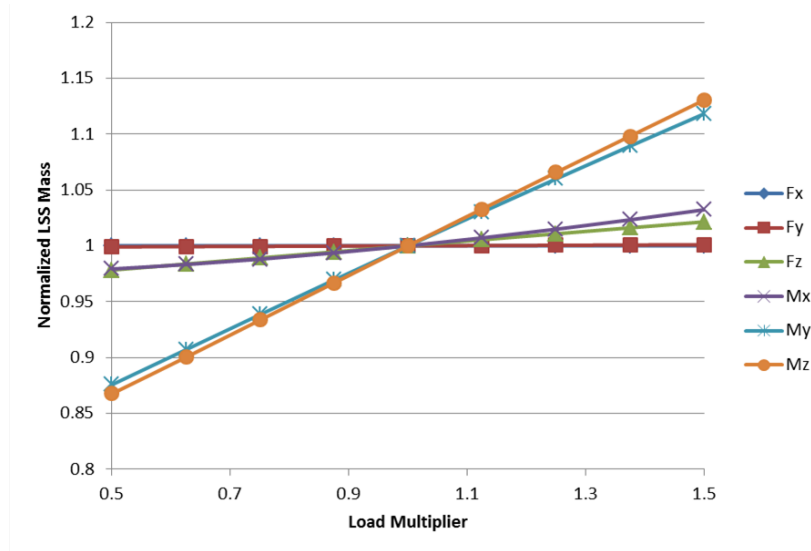


Figure 5. LSS loads sensitivity under four-point configuration

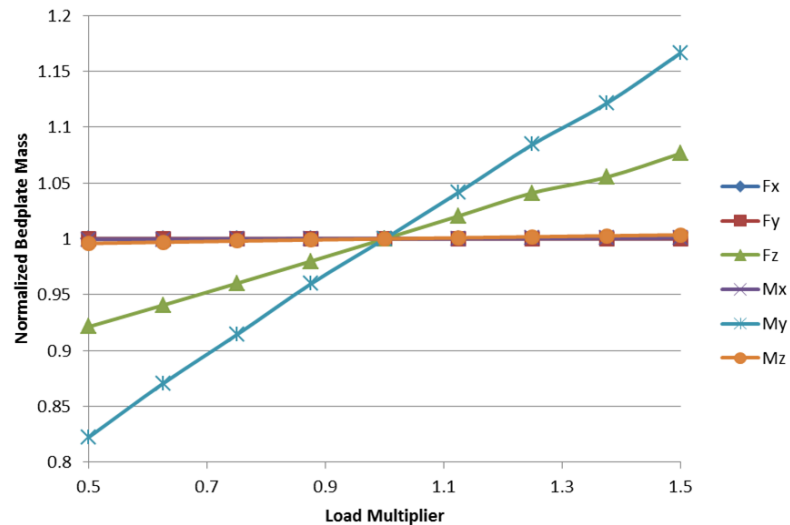


Figure 6. Bedplate loads sensitivity under four-point configuration

Compared to the three-point suspension, the shaft and bedplate design of four-point suspension drivetrains is less sensitive to the rotor loads because of the existence of a second main bearing on the main shaft. Among all rotor loads, M_y and M_z are the key players for both drivetrain configurations. The DLCs that result in large M_y and M_z are the primary design drivers. In other words, the four-point suspension might have a cost benefit over the three-point suspension when the rotor design causes large moments at the hub center, in particular M_y and M_z . Detailed discussion on the weight and cost comparisons between these two drivetrain configurations is presented in the following sections.

4.2. Case Study Mass Results

Table V shows the total nacelle mass from each of the cases and the corresponding masses of the affected components in this study. We find agreement with known masses for the 1.5-MW WindPACT nacelle (52,839 kg) and 5.0-MW NREL reference nacelle (240,000 kg) in land-based conditions under four-point suspensions [16, 17]. Because the reference turbine configurations for these machines include a second main bearing in actuality, these results support the validity of the model and its methods. The respective errors of 7.5% and 4.4% indicate that the model is slightly overpredicting overall

nacelle masses. However, the reference turbines are model-based designs as well and represent single data points. Further model validation by each component and for overall nacelle masses can be found in Guo *et al.* 2015 [3].

Table V. Total nacelle and the corresponding component masses ($\times 10^3$ kg) for each case

Case	Total Nacelle	LSS	Main Bearing	Second Bearing	Gearbox	Bedplate
1	47.2	5.38	3.34	-	19.3	8.65
2	56.8	9.53	2.18	1.74	18.7	12.8
3	216	26.1	12.9	-	60.4	69.6
4	251	33.0	9.54	4.63	59.1	92.7
5	253	38.7	12.9	-	60.7	91.1
6	315	50.9	16.6	6.67	59.1	126

The 5-MW turbine offshore cases 5 and 6 are more massive than their land-based counterparts, cases 3 and 4, due to the higher loads that are due to the coupled dynamics of wind and wave loading. Offshore loads are heavily dependent on local wave spectra as well as the design and damping of the offshore platform. Consequently, we expect component masses for offshore turbines to be higher than those of land-based turbines of the same configuration.

Cases 1, 3, and 5 in Table V show that the three-point designs result in a lower overall nacelle mass than the two-main-bearing designs. The land-based four-point cases result in upwind main bearings that are less massive than the single main bearings in the three-point cases, but the additional mass from the second bearing offsets this benefit. The addition of a second bearing increases the dimensions of the LSS in each case. This is due largely to the extra length needed for the bearings in a four-point main shaft and the difference in the load path downwind of the main bearing. The increase in main shaft and bearing mass is coupled with a decrease in gearbox mass, but because the gearbox is located above the tower-top center in most cases, the resultant loads are transferred directly to the tower. In turn, the bedplates from the four-point cases were designed to be significantly larger than the three-point cases to carry the additional loads and longer moment arms from the shaft and bearings and transfer them to the tower. As a result, the four-point drivetrain configurations are 13% to 20% heavier than three-point drivetrain configurations in an equivalent power rating and operating environment. Note in Table V that the bedplate is the single heaviest component in all configurations, and its mass and dimensions are sensitive to the location and external dimensions of the other nacelle components, resulting in a strong coupling between shaft and bearing system mass and bedplate mass. Cases 2, 4, and 6 have lower gearbox weights than cases 1, 3, and 5. The main differentiator of gearbox weight between three-point and four-point suspension is that the gearbox housing of a three-point suspension is designed to carry the additional nontorque loads transmitted from the wind turbine rotor at the gearbox in addition to torque. The gearbox housing in the four-point suspension is lighter because the nontorque loads at the gearbox are much lower than the three-point suspension. This effect of nontorque loads on gearbox weight is considered in DriveSE by using a scaling factor derived from available industry data. However, the significant weight increase of the rest of the drivetrain components in four-point suspension offsets the benefit of the gearbox weight reduction.

These results show that under the design assumptions within the DriveSE model, and with total mass reduction as the only goal, a three-point suspension configuration would be selected regardless of loading scenario. However, there are higher-level system objectives, such as cost, which may trump these benefits.

4.3. Case Study Cost Comparison

Table VI summarizes the component costs for each case study. In each case, the three-point configuration nacelles are less expensive than the four-point counterparts. From a cost optimization standpoint, the difference between the two design choices is less pronounced than in the mass comparison because of relative cost per-unit-mass across the different components. On average, the nacelle mass decrease from four-point to three-point suspension is 16.7%, while the cost savings between the two averages 3.5%. This is because the majority of the mass savings comes from the bedplate and main shaft, which are less costly on a per-unit-mass basis than the gearbox and bearings. In particular, the gearbox mass is actually reduced in the four-point suspension cases so that there is a cost savings, which counteracts the cost increase for the other components. Using WindPACT data, the turbine cost model finds a \$17-per-kg cost for the gearbox and only a \$1-per-kg cost for the other massive nacelle component, the bedplate [29]. The data used in the model is relatively old, so caution should be taken in interpreting the cost analysis results; however, the materials involved and the complexity in the manufacture of the gearbox relative to the bedplate, LSS and main bearings support the notion that even with today's cost data, the gearbox is still likely to be much more costly than the other drivetrain components. Thus, the lower 3.5% decrease in drivetrain cost compared to 16.7% for drivetrain mass moving from four- to three-point configuration reflects well the impact of the underlying physics in the design on mass of components and how these build up to cost impacts at the system level.

Table VII shows more detail for each reference turbine in terms of the differences in major component costs and overall savings moving from four- to three-point designs. Positive value indicates that a three-point system is cheaper and negative

Table VI. Component costs ($\times 10^6$ \$) and their effects on total nacelle capital cost

Case	Nacelle	LSS	Bearings	Gearbox	Bedplate
1	1.12	0.0499	0.0200	0.413	0.0364
2	1.15	0.0719	0.0235	0.399	0.0419
3	3.56	0.160	0.0772	1.37	0.117
4	3.66	0.196	0.0849	1.34	0.148
5	3.74	0.227	0.0772	1.38	0.146
6	3.96	0.291	0.139	1.34	0.191

value means a three-point system is more expensive. The four-point gearbox cost savings resulting from the model can be readily seen, but these savings are not enough to overcome the increase in the cost of the other components in the nacelle. In the 1.5-MW case, the gearbox savings offsets a large portion of the increase in the other components' costs. However, as the loads increase to the levels of the 5-MW offshore case, the bearings and shaft are significantly more expensive between the two configurations and result in large expected cost savings.

Table VII. Capital cost difference between three- and four-point drivetrains

Turbine Cost Savings ($\times 10^3$ \$)	Total	LSS	Bearings	Gearbox	Bedplate	Other Components
1.5-MW land-based	+26.7	+22.0	+3.48	-14.6	+5.53	+10.2
5.0-MW land-based	+102	+36.8	+7.71	-30.2	+30.7	+56.8
5.0-MW offshore	+218	+64.3	+62.1	-38.7	+45.6	+84.3

Thus, the results would appear to suggest that there is a cost benefit for larger machines (such as General Electric GE 1.X-MW turbines and some GE 2.X-MW turbines) to use a three-point configuration—especially for offshore applications. Despite the apparently lower capital cost for three-point suspension drivetrains, several common wind turbines up to the 5-MW scale have employed a four-point suspension drivetrain design, including the Gamesa G114 2-MW, Vestas V80 2-MW, and General Electric GE 120 2.5-MW turbines. Because wind turbine original equipment manufacturers (OEM) typically make design decisions based on complete system cost considerations that include O&M cost and availability, the results of this study using the simple models in DriveSE don't necessarily match commercial product selections. Future efforts will bring in more fidelity to DriveSE (around fatigue and reliability in particular) so that the model will better capture scenarios where there is an advantage to four-point suspension drivetrains.

There is a more compelling reason why designers would chose a four-point drivetrain when it may be more costly on a turbine capital level. There is a trade-off between reliability and cost benefits for wind turbine design. From the cost-optimization aspect, a three-point suspension drivetrain is superior to a four-point in lowering the capital cost. However, three-point suspension drivetrains are perceived to be less reliable than four-point suspensions. A recent failure database report indicates that gearboxes with three-point suspensions have a failure rate of 5.9% per year, which is more than seven times the 0.8% per year failure rate of those with four-point suspensions [30]. Common failure modes may include gear micropitting, rolling-element bearing contact fatigue, and white-etching cracks on high-speed bearings. Some of the underlying causes for the high failure rates are linked to the nontorque loads experienced at the gearbox. As the DriveSE model shows, the nontorque loads imparted on the gearbox of a four-point suspension turbine are much lower than a three-point suspension. Gearbox reliability has been shown to be a key goal for wind turbine design because of the costly consequences and frequency of gearbox failures [6, 1]. Because DriveSE and the WISDEM cost models do not currently account for the full effects of nontorque loads on a gearbox, or the operational costs associated with drivetrain reliability, the effects of these considerations are missing from the analysis cost results. A rigorous quantitative analysis of the reliability benefits of a four-point versus a three-point drivetrain is outside the scope of this current study. Future development and analysis will aim to improve the fidelity of modeling the effects of fatigue and O&M costs on different drivetrain architectures.

5. CONCLUSIONS

This study used a new physics-based model for sizing wind turbine drivetrain components to explore the capital cost differences in three- and four-point suspension configurations for wind turbines. The bedplate, main bearings, and main shaft are largely sensitive to the nontorque loads from the rotor. The study's mass analysis shows a 16.7% weight savings by using a three-point suspension, mainly caused by the lack of a second bearing, shorter drivetrain length, and lighter

resultant bedplate. The additional bearing causes shaft and bedplate masses to increase substantially, but also results in a comparatively smaller decrease in gearbox mass. As a result of the mass differences, significant cost savings are seen in the main shaft and bedplate of three-point suspension drivetrains. Because gearboxes are more expensive than the main shaft and bedplate on a per-unit-mass basis, the percentage cost savings is less than the percentage mass savings. Three-point suspension turbines cost an average of 3.5% less in terms of capital cost than their corresponding four-point versions. This analysis, however, does not consider O&M cost associated with drivetrain reliability. Designers need to consider both cost optimization and reliability perspectives to achieve their best design practice.

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REFERENCES

1. R. Bergua, J. Jove, J. Echarte, Pure torque drivetrain design: A proven solution for increasing the wind turbine reliability, Tech. rep., ALSTOM Wind S.L.U. (2014).
2. H. Link, W. LaCava, J. van Dam, B. McNiff, S. Sheng, R. Wallen, M. McDade, S. Lambert, S. Butterfield, F. Oyague, Gearbox reliability collaborative project report: findings from phase 1 and phase 2 testing, Tech. Rep. NREL/TP-5000-51885, National Renewable Energy Laboratory (2011).
3. Y. Guo, T. Parsons, R. King, K. Dykes, P. Veers, An analytical formulation for sizing and estimating the dimensions and weight of wind turbine hub and drivetrain components, Tech. Rep. NREL/CP-5000-63008, National Renewable Energy Laboratory (2015).
4. J. Coultate, Understanding costs for large wind-turbine drivetrains, <http://www.windpowerengineering.com>, accessed: 2015-05-05.
5. Y. Guo, J. van Dam, R. Bergua, J. Jove, J. Campbell, Improving wind turbine drivetrain reliability using a combined experimental, computational, and analytical approach, Tech. Rep. NREL/CP-5000-61683, National Renewable Energy Laboratory (2015).
6. F. Oyague, Gearbox reliability collaborative project 750 / 48.2 description and loading document (iec 61400-1 class iib), Tech. rep., NREL (2010).
7. G. Jenkins, R. Harrison, Cost modeling of horizontal axis wind turbines, Tech. Rep. ETSU/W-34-00170-REP, University of Sunderland School of Environment (December 1993).
8. L. Fingersh, M. Hand, A. Laxson, Wind turbine design cost and scaling model, Technical Report NREL/TP-500-40566, National Renewable Energy Laboratory (Dec. 2006).
URL <http://www.osti.gov/bridge/servlets/purl/897434-Feoadi/>
9. Wind partnerships for advanced component technology, <http://www.nrel.gov/wind/windpact.html> (May 2013).
10. K. Dykes, WisdemTM, <http://nwtc.nrel.gov/WISDEM>, accessed: 2015-04-20 (2014).
11. A. Ning, K. Dykes, Understanding the benefits and limitations of increasing maximum rotor tip speed for utility-scale wind turbines, Journal of Physics: Conference Series 524 (1) (2014) 012087.
URL <http://stacks.iop.org/1742-6596/524/i=1/a=012087>
12. K. Dykes, B. Resor, A. Platt, Y. Guo, R. Ning, A. and King, T. Parsons, D. Petch, P. Veers, Effect of tip-speed constraints on the optimized design of a wind turbine, Tech. Rep. NREL/CP-5000-61726, National Renewable Energy Laboratory (2014).
13. AGMA, Ansi/agma 6001-e08: design and selection of components for enclosed gear drives, Tech. rep., AGMA (2008).
14. J. Shigley, C. Mischke, R. Budynas, Mechanical Engineering Design, 7th Edition, McGraw-Hill Science/Engineering/Math, Boston, 2003.
15. K. Dykes, Drivewpact 0.1.0 documentation, <http://wisdem.github.io/DriveWPACT>, accessed: 2015-02-15 (2014).
16. J. Jonkman, S. Butterfield, W. Musial, G. Scott, Definition of a 5-MW reference wind turbine for offshore system development, Technical Report NREL/TP-500-38060, National Renewable Energy Laboratory (Feb. 2009).
URL <http://www.osti.gov/bridge/servlets/purl/947422-nhrlni/>
17. D. J. Malcolm, A. C. Hansen, WindPACT turbine rotor design study, Subcontract Report NREL/SR-500-32495, National Renewable Energy Laboratory (April 2006).
URL <http://www.nrel.gov/docs/fy06osti/32495.pdf>

18. S. Frandsen, N. J. Tarp-Johansen, E. Norton, K. Argyriadis, B. Bulder, K. Rossis, , Tech. Rep. ENK5-CT-2000-00322 (March 2005).
URL [http://www.risoe.dk/vea/RECOFF/Internal/ProgressReports/Final%20technical%20report%20RECOFF%20\(contract%20no%20ENK5-CT-2000-00322\).pdf](http://www.risoe.dk/vea/RECOFF/Internal/ProgressReports/Final%20technical%20report%20RECOFF%20(contract%20no%20ENK5-CT-2000-00322).pdf)
19. H. J. T. Kooijman, C. Lindenburg, D. Winkelaar, E. L. van der Hooft, DOWEC 6 MW pre-design, Tech. Rep. DOWEC-FIW2-HJK-01-046/9, Energy Research Center of the Netherlands (ECN) (Sep. 2003).
URL https://www.ecn.nl/fileadmin/ecn/units/wind/docs/dowec/10046_009.pdf
20. J. M. Jonkman, Dynamics of offshore floating wind turbines-model development and verification, *Wind Energy* 12 (5) (2009) 459–492. doi:10.1002/we.347.
URL <http://doi.wiley.com/10.1002/we.347>
21. B. R. Resor, Definition of a 5MW/61.5m wind turbine blade reference model, Sandia Report SAND2013-2569, Sandia National Laboratories (Apr. 2013).
URL <http://prod.sandia.gov/techlib/access-control.cgi/2013/132569.pdf>
22. M. J. Churchfield, S. Lee, J. Michalakes, P. J. Moriarty, A numerical study of the effects of atmospheric and wake turbulence on wind turbine dynamics, *Journal of Turbulence* (2012) N14doi:10.1080/14685248.2012.668191.
URL <http://www.tandfonline.com/doi/abs/10.1080/14685248.2012.668191>
23. J. Park, S. Basu, L. Manuel, Large-eddy simulation of stable boundary layer turbulence and estimation of associated wind turbine loads, *Wind Energy* 17 (3) (2014) 359–384. doi:10.1002/we.1580.
URL <http://onlinelibrary.wiley.com/doi/10.1002/we.1580/abstract>
24. S. Lee, M. Churchfield, P. Moriarty, J. Jonkman, J. Michalakes, Atmospheric and wake turbulence impacts on wind turbine fatigue loadings, American Institute of Aeronautics and Astronautics, 2012. doi:10.2514/6.2012-540.
URL <http://arc.aiaa.org/doi/abs/10.2514/6.2012-540>
25. Y. Bazilevs, M. C. Hsu, M. A. Scott, Isogeometric fluidstructure interaction analysis with emphasis on non-matching discretizations, and with application to wind turbines, *Computer Methods in Applied Mechanics and Engineering* 249 (2012) 28–41. doi:10.1016/j.cma.2012.03.028.
URL <http://www.sciencedirect.com/science/article/pii/S0045782512001041>
26. J. M. Jonkman, M. L. Buhl Jr, FAST user's guide, Golden, CO: National Renewable Energy Laboratory.
URL <https://wind.nrel.gov/designcodes/simulators/fast/FAST.pdf>
27. G. Hayman, A matlab-based generator for extreme-event tables, <https://nwtc.nrel.gov/MEXtremes>, accessed: 2015-02-15 (2015).
28. SKF, Bearing product tables, <http://www.skf.com/us/products/bearings-units-housings/product-tables/index.html>, accessed: 2014-09-29 (2014).
29. K. Dykes, Turbine costs 0.1.0 documentation, accessed: 2015-02-15 (2014).
URL <http://wisdem.github.io/TurbineCostsSE>
30. B. Berger, Major component failure data & trends, Operations and Maintenance Summit - Proceedings, Toronto, ON, 2016.

APPENDIX: DLCS USED IN CASE STUDY ANALYSIS

The results below provide a summary of the extreme loads from a select set of IEC 61400-1 and 61400-3 design load cases for each turbine case used in the study. The DLCs associated with power production, power production plus occurrence of fault, parked, and parked with fault have been studied. Each DLC has been run multiple times with different wind profiles. These tables list the forces and moments in all three axes within the nonrotating coordinate system. The moments are defined at the tower top location. The values in these tables are the maximum values with consideration of the safety factors defined in IEC 61400-1 and 61400-3 design standards. A brief description of the selected DLCs is given in Table VIII. For each row, the instantaneous values of all other forces and moments are provided when individual shaft loads reach the maximum or minimum. The design load cases with the extreme shaft loads are also listed.

Table VIII. IEC DLC cases

DLC	Meaning
1.1	Power production
1.3	Power production with extreme turbulence conditions
2.3	Extreme operating gust combined with electrical system fault
6.1	Parked with extreme wind speed (50-year recurrence)
6.2	Parked with extreme wind speed and loss of electrical network connection (50-year recurrence)
7.1	Parked with extreme wind speed (1-year recurrence)

Table IX. WindPact 1.5-MW Extreme Loads Table

Load Case	Type	Thrust (kN)	LSS Fy (kN)	LSS Fz (kN)	Torque (kNm)	LSS My (kNm)	LSS Mz (kNm)
DLC 2.3	Minimum	-155.994	-3.400	-284.130	-231.330	-248.930	-334.620
DLC 1.1	Maximum	448.889	3.249	-407.028	1,173.450	-706.950	661.500
DLC 1.3	Minimum	91.732	-77.703	-332.640	949.590	1,336.223	-918.405
DLC 1.3	Maximum	130.275	68.330	-355.320	921.611	-1,902.150	-597.645
DLC 1.1	Minimum	147.375	-13.292	-461.946	1,091.194	119.260	1,542.000
DLC 2.3	Maximum	46.617	-1.951	-204.366	-99.847	-165.000	-281.380
DLC 2.3	Minimum	-56.232	-5.345	-244.200	-270.294	-148.339	-643.720
DLC 1.1	Maximum	185.250	-15.285	-405.956	1,340.390	741.580	396.150
DLC 1.3	Minimum	262.043	28.026	-347.625	843.885	-2,675.768	752.220
DLC 1.3	Maximum	153.090	-57.537	-353.025	1,053.540	2,129.116	-283.365
DLC 1.3	Minimum	306.990	-1.293	-332.100	1,048.005	-241.380	-2,273.131
DLC 1.3	Maximum	273.914	8.733	-403.110	894.240	556.740	2,650.607

Table X. NREL 5-MW reference turbine (Land-Based)

Load Case	Type	Thrust (kN)	LSS Fy (kN)	LSS Fz (kN)	Torque (kNm)	LSS My (kNm)	LSS Mz (kNm)
DLC 2.3	Minimum	-894.190	1.906	-1,166.000	-1,423.400	-1,675.300	-1,384.900
DLC 1.3	Maximum	1,506.600	-3.178	-1,474.200	6,115.500	-537.975	902.070
DLC 6.2	Minimum	-5.222	-1,093.950	-855.690	1,124.200	-8,072.900	6,613.200
DLC 6.2	Maximum	78.848	1,119.800	-864.160	-1,035.980	1,078.550	-3,135.000
DLC 1.1	Minimum	581.400	-27.045	-1,761.000	6,177.000	-1,794.000	4,776.000
DLC 6.1	Maximum	31.887	-465.480	-313.335	90.760	-1,786.050	6,980.850
DLC 2.3	Minimum	-580.800	-87.076	-1,049.070	-2,253.900	2,063.600	-5,431.800
DLC 1.1	Maximum	647.700	-8.625	-1,454.400	7,836.000	313.350	-9,579.000
DLC 7.1	Minimum	599.610	186.780	-842.710	330.770	-16,665.000	2,896.300
DLC 1.3	Maximum	254.475	-179.145	-1,364.850	4,942.350	14,053.500	-5,404.050
DLC 1.3	Minimum	300.510	43.227	-1,238.085	4,326.750	-1,055.160	-14,607.000
DLC 1.3	Maximum	1,362.150	40.257	-1,513.350	6,347.700	-3,508.650	13,132.800

Table XI. NREL 5-MW reference turbine (Offshore)

Load Case	Type	Thrust (kN)	LSS Fy (kN)	LSS Fz (kN)	Torque (kNm)	LSS My (kNm)	LSS Mz (kNm)
DLC 1.1	Minimum	-2,780.000	1,530.000	-938.000	3,470.000	-16,400.000	-3,460.000
DLC 1.1	Maximum	4,400.000	-424.000	-1,100.000	8,540.000	-2,120.000	10,700.000
DLC 1.1	Minimum	-1,180.000	-2,070.000	265.000	4,430.000	5,500.000	6,480.000
DLC 1.1	Maximum	-2,540.000	2,150.000	-107.000	3,160.000	-4,650.000	-12,500.000
DLC 1.1	Minimum	-1,540.00	143.000	-2,080.000	3,360.000	-10,200.000	7,760.000
DLC 1.1	Maximum	-602.000	-169.000	2,000.000	4,390.000	-6,490.000	-1,090.000
DLC 1.1	Minimum	-1,920.000	-1,350.000	1,060.000	-1,650.000	-16,900.000	5,550.000
DLC 1.1	Maximum	2,490.000	366.000	-1,310.000	10,700.000	2,140.000	-1,250.000
DLC 1.3	Minimum	488.000	492.000	822.000	4,500.000	-29,100.000	15,800.000
DLC 1.1	Maximum	43.300	426.000	-821.000	2,070.000	32,600.000	7,210.000
DLC 1.3	Minimum	1,610.000	-996.000	456.000	6,510.000	-5,110.000	-27,700.000
DLC 1.1	Maximum	-1,500.000	-1,020.000	-987.000	5,100.000	1,530.000	26,200.000